

$$M = J\beta = 63,6 \cdot 1,52 \approx 100 \text{ k.m}$$

Orden: 100 k.m

Dz am 12.10
~ 1(2.1)

Dano:

$$T = 5 \frac{\text{seg}}{\text{C}}$$

$$E_k = 60 \text{ Dne}$$

L - ?

Seu nome:

$$T = 5 \frac{\text{seg}}{\text{C}} = 10\pi \frac{\text{seg}}{\text{C}}$$

$$L = J\omega; E_k = \frac{J\omega^2}{2}$$

$$2E_k = J\omega^2, \frac{2E_k}{\omega} = J\omega$$

$$\Rightarrow L = 2 \frac{E_k}{\omega}$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{10\pi} = \frac{1}{5}$$

$$L = 2E_k \cdot \frac{1}{\omega} = 2 \cdot 60 \cdot 5 = 600$$

Orden: 600 $\frac{\text{m}^2 \cdot \text{kg}}{\text{C}}$

Dano:

$$M_3 = 5,926 \cdot 10^{24} \text{ kg}$$

$$R_3 = 6,327 \cdot 10^6 \text{ m}$$

L - ?

~ 2(7.2)

Seu nome:

$$L = J\omega = \frac{2}{5} M_3 R_3^2 \omega$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{24 \cdot 60 \cdot 60} =$$

$$= \frac{2\pi}{24 \cdot 60 \cdot 60}$$

$$\Rightarrow L = \frac{2}{5} \cdot 5,926 \cdot 10^{24} \cdot 6,321^2 \cdot 10^{12} \cdot \frac{\pi}{24 \cdot 60 \cdot 30} \approx$$

$$\approx 2,056 \cdot 10^{33} \frac{\text{m}^2 \text{kg}}{\text{s}}$$

$$\text{Geben: } 2,056 \cdot 10^{33} \frac{\text{m}^2 \text{kg}}{\text{s}}$$

~ 3. (12.3)

Teilelement:

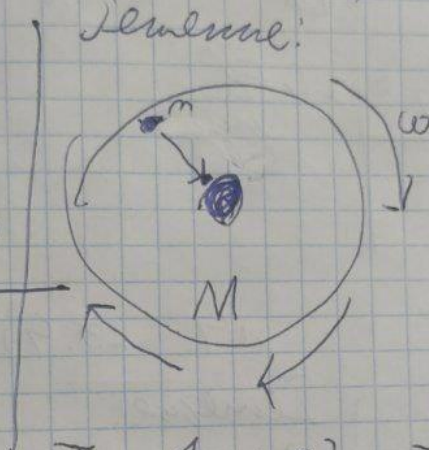
Dann:

$$m = 60 \text{ kg}$$

$$M = 120 \text{ kg}$$

$$v_1 = 12 \frac{\text{km}}{\text{h}}^{-1}$$

$$\Rightarrow v_2 = ?$$



$$v_1 = 12 \frac{\text{km}}{\text{h}}^{-1} = 2 \frac{\text{km}}{\text{h}}^{-1} = 2 \cdot 10^{-3} \frac{\text{m}}{\text{s}}$$

$$\Rightarrow T_1 = \frac{1}{0,2} \text{ s} = 5 \text{ s}$$

$$J_M = \frac{1}{2} M R^2; J_m = m R^2$$

$$J_{\text{Ges}} = J_M + J_m = R^2 \left(\frac{1}{2} M + m \right) = 120 R^2$$

$$M_1 = F_1 R = m g R; M_2 = 0$$

$$\omega_1 = 2\pi \nu = 2\pi \cdot 0,2 = 0,4\pi \frac{\text{rad}}{\text{s}}$$

m.k. werden reduziert & gleich, da $R' = 0$:

$$J'_{\text{Ges}} = J_M = 60 R^2$$

Im 3. Gg. kommt noch was:

$$J \omega_1 = J \omega_2;$$

$$120 R^2 \cdot \frac{2\pi}{5} = 60 R^2 \cdot \omega_2 \quad | : R^2$$

$$120 \cdot \frac{2\pi}{5} = 60 \cdot \omega_2$$

$$\omega_2 = \frac{120 \cdot \frac{2\pi}{5}}{60} = \frac{4\pi}{5} = \frac{4}{5} \pi \text{ rad/s}$$

$$\Rightarrow V_2 = \frac{\omega_2 \cdot 60}{2\pi} = \frac{4\pi}{5} \cdot \frac{30}{\pi} = 24 \frac{\text{cm}}{\text{s}}$$

Answer: $V_2 = 24 \frac{\text{cm}}{\text{s}}$

Demo:

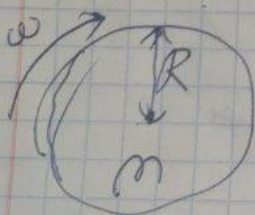
$$m = 20 \text{ kg}$$

$$R = 0,6 \text{ m}$$

$$\mu = 0,4$$

$$E = 10000 \text{ J}$$

$L = ?$



Demo:

$$L = J\omega; J = \frac{1}{2} m R^2 =$$

$$= \frac{1}{2} \cdot 20 \cdot 0,36 = 3,6 \text{ kg} \cdot \text{m}^2$$

$$E_K = \frac{J\omega^2}{2}; \omega = \sqrt{\frac{2E_K}{J}} = \sqrt{\frac{2\mu E}{J}}$$

$$\Rightarrow L = 3,6 \sqrt{\frac{2 \cdot 0,4 \cdot 10000}{3,6}} \approx$$

$$\approx 769,2 \frac{\text{m}^2 \text{ kg}}{\text{s}}$$

Answer: $769,2 \frac{\text{m}^2 \text{ kg}}{\text{s}}$

Demo:

$$J = 4$$

$$M = 2$$

$$V_0 = 0$$

$$t = 1$$

$$F_K =$$

Dano:

$$J = 40 \text{ kg} \cdot \text{m}^2$$

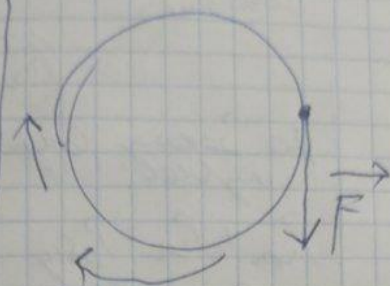
$$M = 20 \text{ N} \cdot \text{m}$$

$$v_0 = 0$$

$$t = 10 \text{ s}$$

$$F_k = ?$$

15 (3.46)
Zadanie:



$$E_k = \frac{J\omega^2}{2} = \frac{1}{2} m R^2 \omega^2 = \frac{m R^2 \omega^2}{4}$$

$$L = J\omega = \frac{1}{2} m R^2 \omega$$

$$M = RF; \omega = \beta t, \omega_m$$

rychle $\beta < 0$

~~logarytm $R = R = \frac{M}{F}$~~

~~$E_k = \frac{m R^2 \omega^2}{4} = \frac{m \frac{M^2}{F^2} \omega^2}{4}$~~

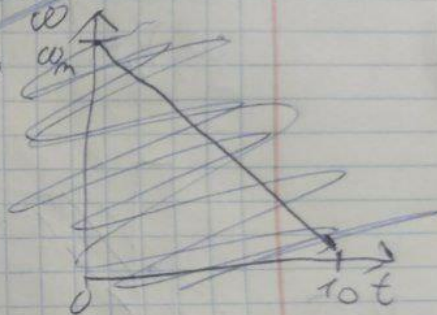
$$\varphi = \frac{\beta t^2}{2}; M = \beta J \Rightarrow \beta = \frac{M}{J}$$

$$\text{wtedy } \varphi = \frac{\frac{20}{40} \cdot 10^2}{2} = 25 \text{ rad}$$

$$\omega = \beta t = \left(\frac{20}{40} \right) \cdot 10 = 5 \frac{\text{rad}}{\text{s}}$$

$$E_k = \frac{J\omega^2}{2} = \frac{40 \cdot 25}{2} = 500 \text{ J}$$

Odpowiedź: 500 J



26. (23.29.)

Dado:

$$m_1 = 0,2 \text{ kg}$$

$$L = 1 \text{ m}$$

$$V_{in} = 10 \frac{\text{m}}{\text{s}}$$

$$m_2 = 10 \text{ g}$$

$$AO = \frac{L}{2}$$

$$\omega, V_K = ?$$

Señaleme:

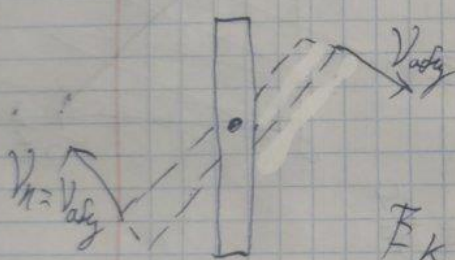
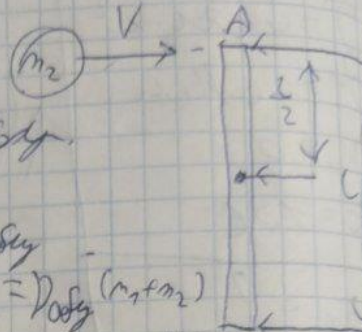
10 Baxany Galy.
unyylcel:

$$P_{cm} + P_{cu} = P_{osy}$$

$$m_1 \cdot 0 + m_2 V_{in} = P_{osy} (m_1 + m_2)$$

$$\Rightarrow 0,01 \cdot 10 = (0,01 + 0,2) V_{osy}$$

$$V_{osy} = \frac{0,1}{0,21} = \frac{10}{21} \frac{\text{m}}{\text{s}} \approx 0,48 \frac{\text{m}}{\text{s}}$$



$$F_{Ku} = \frac{m_2 V_{in}^2}{2} = \frac{0,01 \cdot 100}{2} = 0,5 \text{ J}$$

$$F_K = \frac{J \omega^2}{2} = \frac{\frac{1}{2} m_1 L^2 \omega^2}{2} =$$

$$\frac{0,2 \cdot 1 \cdot \omega^2}{2} = \frac{0,2 \omega^2}{2} = \frac{\omega^2}{10}$$

$$F_K = \frac{m_2 V_{in}^2}{2} = \frac{0,01 \cdot 100}{2} = \frac{0,1 \cdot 100}{2} = \frac{10}{21^2}$$

$$\Rightarrow \frac{\omega^2}{10} = \frac{10}{21^2}; \omega^2 = \frac{100 \cdot 10}{21^2}; \omega = \sqrt{\frac{1000}{21^2}} =$$

$$\approx \frac{10\sqrt{10}}{21} \approx 1,65 \frac{\text{m}}{\text{s}}$$

Onben:

Dado:

$$L = 0,3$$

$$d = 0,1$$

$$T = ?$$

$$L = ?$$

$$T =$$

Onben:

Dado:

$$R =$$

$$\frac{AB}{BC}$$

$$L, T$$

Onben: $V = 0,48 \frac{\text{m}}{\text{s}}; \omega = 1,65 \frac{\text{m}}{\text{s}}$